IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-029

1. Topic (12 words max)

Electric Vehicle Management System Using Artificial Intelligence and Machine Learning

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

The rapid pace of electric vehicle (EV) development is facilitating sustainable transportation and reducing emissions worldwide [1]. Despite advances in EV hardware, battery technologies, and connectivity, many EV operational intelligence and user support systems are still reactive and fragmented. Existing EV systems lack predictive, behavioral, or context-based decision support, requiring only essential metrics such as battery charge, simple warning notifications, and charging station location to be monitored. As a result, important aftermarket functions such as battery health degradation prediction [2], intelligent fault diagnosis [3], predictive charging station location [4][5], breakdown assistance, and performance analytics are not efficiently enabled by built-in real-time analytics.

Nature and Gravity of the Problem

This issue affects multiple stakeholders, such as EV owners, manufacturers, sellers, after-sales service providers, and charging infrastructure operators. EV customers find it difficult to understand notifications, learn about battery degradation [6][7], and make rational choices in the event of a failure or charging overload. Meanwhile, manufacturers lack structured data on how well the product is being used after the sale and cannot determine failure patterns, battery stress patterns, or region-specific operational issues, which means limited opportunity to adjust the product and customer experience.

Impact of the Problem:

Lack of clarity in fault reporting and unreliable EV performance reduce user trust. Early battery degradation, poor charging practices, and poor warranty procedures lead to lost money. Lack of predictive analytics leads to reactive maintenance [8][9]. Inefficient energy consumption and short battery life increase electronic waste and resource drain, and lack of battery health transparency reduces manufacturer accountability [10].

References:

1. International Energy Agency (IEA), Global EV Outlook 2023, Paris, France: IEA, 2023.
2. Y. Zhang, R. Xiong, H. He, and M. Pecht, “Battery degradation modeling for electric vehicles,” IEEE Transactions on Industrial Electronics, 2018.
3. X. Hu, J. Jiang, D. Cao, and B. Egardt, “Battery health prediction and fault diagnosis for electric vehicles,” IEEE Transactions on Vehicular Technology, 2020.
4. S. Wang, B. Liu, Q. Li, D. Han, and J. Zhou, “Electric vehicle charging behavior analysis and load prediction,” Sustainability, 2025.
5. S. Shaheen and A. Cohen, “Smart charging strategies and challenges for electric vehicle infrastructure,” Energy Policy, 2020.
6. M. Berecibar, I. Gandiaga, I. Villarreal, N. Omar, J. Van Mierlo, and P. Van den Bossche, “State of health estimation of lithium-ion batteries: Advances and challenges,” Renewable and Sustainable Energy Reviews, vol. 56, pp. 572–587, Apr. 2016.
7. R. Xiong, F. Sun, and Z. Chen, “Battery degradation and performance assessment of lithium-ion batteries,” Journal of Power Sources, 2014.
8. A. Farmann, “Model-based diagnosis of battery systems for electric vehicles,” Journal of Energy Storage, 2015.
9. Y. Zheng, “Machine learning-based fault detection for electric vehicle batteries," Applied Energy, vol. 300, p. 117346, 2021.
10. Y. Tian, G. Dai, and T. Yan, “Data-driven prediction of lithium-ion battery thermal behavior,” Processes, 2025.

6. Brief description of Existing Research and Systems (300 words max)

Existing Products/Systems

Currently, the electric vehicle (EV) infrastructure consists of charging stations, advanced energy management systems, and Vehicle-to-Grid (V2G) technology, which allows for the flow of electricity between EVs and the electric power system [1]. Smart chargers utilize information and operational methods to maximize efficiency, decrease costs, and minimize strain on the electrical grid [2]. V2G technology utilizes batteries in EVs to function as a decentralized provider of electricity, which supports the electric power grid [3]. These technologies will allow for the increased incorporation of renewable energy resources into the electric power grid and improve the overall resilience of the electrical energy system.

Existing Research

Current EV studies are conducted under the heading of EV infrastructure, grid interaction and charging demand forecasting [1]. Emerging technologies involved in variable load improvement, better power supply and supporting grid services are smart charging and V2G as the main solutions considered by the studies [2][3]. The combination of EV charging and smart grids and energy internet to improve data communication and coordinated energy management of both renewable and non-renewable sources is also studied. Demand forecasting models are data-driven methods that improve network planning and renewable energy use [4]. All these studies reflect the importance of smart systems to control EV development without negatively affecting power systems.

Gaps/Limitation identified

Despite the progress, there are significant limitations. Most studies are based on simulated or static data, which limits their application to the real world [4]. Current forecasting models have difficulty capturing the temporal and spatial variability of renewable generation, user behavior, and weather conditions. Moreover, detailed frameworks between charging demand and infrastructure planning, socio-economic factors, and policies have yet to be developed.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

I. Nature of the Proposed Solution

The proposed solution is an Integrated Intelligent Decision Support System (I-IDSS) for Electric Vehicles (EVs), designed to address critical limitations identified in existing post-sale EV management platforms. Unlike conventional EV systems that primarily focus on real-time monitoring and reactive alerts, the proposed I-IDSS leverages artificial intelligence, predictive analytics, and user-centric decision support to deliver proactive, behavior-based, and context-aware insights. The core objective of the system is to enhance post-sale performance evaluation, failure prediction, and decision-making for both EV users and EV sellers through a unified, data-driven architecture.

Addressing Gaps Identified in Existing Systems

Most current EV applications lack advanced analytical capabilities such as post-sale performance evaluation, behavioral analytics, trend-based failure prediction, and contextual user guidance. As a result, EV users receive limited assistance beyond basic diagnostics, while EV sellers have minimal visibility into how vehicles perform under real-world usage conditions. This information gap restricts sellers from making informed decisions related to vehicle design improvements, warranty policies, and service optimization.

The proposed I-IDSS overcomes these limitations by introducing a multi-stakeholder system architecture that integrates anonymized user-generated vehicle data with advanced analytical models. A dedicated Performance Intelligence and Failure Prediction (PIFP) module is provided for EV sellers, enabling the identification of recurring failure patterns, battery stress conditions, and region-specific performance issues. These insights support data-driven improvements in product quality, after-sales services, and operational strategies.

For EV users, the system extends beyond traditional dashboards by offering predictive battery health analysis, intelligent fault diagnosis, and personalized guidance. By analyzing individual driving behaviors and charging patterns, the system generates customized battery degradation forecasts and explains the underlying causes of performance decline. Additionally, the context-aware guidance engine provides actionable “do’s and don’ts” in response to warnings, reducing user anxiety and preventing further vehicle damage.

II. System Perspective: Input, Process, Output

Inputs to the system include EV usage behavior data (charging frequency, charging type and duration, trip distance, and driving stress patterns), sensor and battery data (temperature anomalies, electrical alerts), user-reported breakdowns, charging station usage statistics, and location-based data such as nearby service stations and charging routes.

Processing is carried out through multiple intelligent modules. The post-sale performance intelligence engine applies machine learning techniques to analyze aggregated anonymized data and identify failure trends and performance anomalies. The adaptive battery health prediction module estimates battery lifespan based on user-specific driving and charging behaviors. An intelligent diagnosis engine combines rule-based reasoning and predictive models to assess fault severity and urgency, while the predictive charging and breakdown assistance module forecasts charger availability and provides optimized charging and emergency routing recommendations.

Outputs include seller-oriented dashboards presenting post-sale performance insights and failure trends, user-focused battery health timelines, context-aware fault alerts with actionable guidance,

predictive charging station recommendations, breakdown route assistance, and real-time decision support for both EV users and sellers.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

- **Access to Specific Domain Expertise**

The proposed research requires multidisciplinary expertise across electric vehicle (EV) systems, battery health management, charging infrastructure, and aftermarket vehicle performance analysis. Essential field knowledge will be obtained through a systematic literature review of peer-reviewed journals, conference proceedings, and technical reports focusing on EV battery degradation, charging behavior, vehicle diagnostics, and intelligent transportation systems. These scholarly sources provide scientifically valid insights into EV operating characteristics and common real-world challenges faced in EV use.

In addition, technical documents published by EV manufacturers, energy research organizations, and International Standards Development Organizations (ISDOs) will be reviewed to obtain practical guidance on charging standards, battery safety limits, and operational limitations. Academic supervision and expert advisory support will further enhance domain understanding, ensuring alignment between research objectives and real-world EV performance needs.

- **Developing Technical Knowledge and Research Skills**

A strong foundation in machine learning, predictive analytics, and intelligent decision support system design is central to this research. Team members will develop the necessary technical skills through coursework, structured self-study, and experimental implementation. Basic machine learning techniques including supervised classification, regression analysis, and time series forecasting will be explored through hands-on experiments using open-source data science frameworks.

Key research skills such as feature engineering, model evaluation, error analysis, and result interpretation are emphasized to maintain methodological rigor. These skills will be complemented by software engineering knowledge related to system architecture, API integration, database management, and dashboard development. This integrated skill set supports the development of a comprehensive research prototype that can deploy analytical models via mobile and web-based interfaces, thereby enhancing the practical relevance of research results.

- **Data Availability and Collection Strategy**

Multiple data acquisition strategies will be used, including simulation-based data generation, use of publicly available datasets, and voluntary data sharing by participants through research-oriented mobile applications. The data collected will be non-intrusive and anonymous, focusing on EV operational metrics such as charging events, battery performance degradation indicators, and breakdown or failure patterns.

In cases where real-world datasets are limited – particularly for charging infrastructure availability and failure scenarios – simulated datasets will be included to support broader analysis. To ensure transparency and reproducibility, data extraction and processing methodologies will be documented and published via a publicly accessible platform. Robust anonymization and collection mechanisms will be implemented to protect participant privacy while enabling future research use.

- **Ethical Considerations and Clearance Requirements**

Ethical considerations arise due to the participation of human participants who contribute to vehicle usage data. The study relies on informed consent, secure data handling, and anonymization techniques, and does not require the collection of sensitive personal identifiers. The risk to the participant is minimal.

Although ethical approval may not be strictly mandatory due to the voluntary, anonymous, and non-invasive nature of data collection, obtaining clearance from an institutional ethics review committee reinforces compliance with research governance standards and enhances the credibility and acceptability of study findings.

9. Objectives and Novelty

Main Objective

The purpose of this research is to create an Integrated AI-Driven Smart Decision Support System for Electric Vehicles (EVs), which will help improve the reliability of EVs after purchase and the enjoyment of EVs by predicting the deterioration of battery health, diagnosing functional problems with the vehicle, suggesting intelligent ways to charge the vehicle, and providing assistance for breakdowns, thereby increasing accuracy, safety, and sustainability related to EV decision-making for EV users and manufacturers.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22157928	AI-Based EV Anomaly Detection	• Collect anonymous after-sales EV usage data from users (battery, charging, errors). • Analyze real-world performance trends using analytics/ML. • Identify frequent failures and battery stress patterns. • Generate dashboards for sellers (issues by region, by model). • Support warranty and service decision-making.	This component shifts EV research focus from *selling* to *post-sale intelligence* by using real user data to predict failures and improve vehicle design and after-sales service.
IT22247490	Adaptive Battery Health Prediction Based on Charging & Driving Behavior	• Monitor charging habits (fast/slow charging). • Track driving behavior (speed, acceleration, trip length). • Predict future battery health using ML	This component introduces a behavior-based battery health prediction by predicting future battery degradation using individual charging and driving patterns, rather

		models. • Display battery health trends (3, 6, 12 months). • Explain how user behavior affects battery degradation.	than relying solely on the current battery percentage.
IT22134080	Intelligent EV Charging Optimization and Range Prediction	• Detect EV alerts and abnormal sensor data (DL: LSTM / Autoencoder) • Classify errors by severity (ML: Random Forest / XGBoost) • Generate easy-to-follow dos and don'ts (Rule-based + NLP) • Recommend emergency actions (DL + Rule-based Logic) • Show clear alerts to reduce user confusion (UI + Visualization)	This component transforms technical EV error data into human-understandable guidance, helping users determine what action to take instead of displaying raw error codes.
IT22071170	AI-Based EV Virtual Assistant Using Retrieval Augmented Generation (RAG) with Voice Support	• Predict charging station slot availability using past usage data. • Recommend optimal charging stations with wait time estimates. • Identify or acknowledge breakdown alerts from users. • Calculate battery-safe routes to service stations. • Provide emergency driving and safety advice.	This component introduces predictive and emergency-aware support by combining charge availability prediction with intelligent breakdown routing based on battery status.

SLIIT UNT
THE KNOWLEDGE UNIVERSITY

IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Approved.

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Uthpala	Samarakoon	14/01/2026
Co-Supervisor	Mr.	Nelum	Amarasena	14/01/2026
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				